

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 4.2: Probability I — Teacher Reference (Pre-filled)

SUMMARY TABLE: MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 4.2: Probability I
Driving Question	DRIVING QUESTION: How can we use mathematics to measure and predict the likelihood of uncertain everyday events — like Mama Njeri's mandazi sales — so we can make smarter decisions even when we cannot be certain of the outcome?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Mandazi Stall Mystery: Introducing Uncertainty and Experimental Probability	Students observed three parallel data sets — Mama Njeri's 60-day sales record, coin-flip results, and bottle-cap toss results — and plotted running relative frequencies for all three. They noticed that relative frequencies fluctuate early but appear to stabilise toward a consistent value as the number of trials increases.	Students learned that probability is a mathematical measure of likelihood ranging from 0 (impossible) to 1 (certain); that a random experiment has an uncertain outcome and a defined sample space; that experimental probability is calculated as the number of times an event occurs divided by the total number of trials; and that experimental probability becomes more reliable (stabilises) as the number of trials increases.	Students explained the pattern in Mama Njeri's notebook and their own experiments using the concept of experimental probability. They described why individual days are unpredictable yet the fraction of good days stabilises over 60 days — because with more trials, random variation averages out and the relative frequency converges toward the true probability of a good day. This explains why keeping a longer notebook makes Mama Njeri's data more mathematically useful for planning.
Lesson 2 Experimental Probability II: More Trials, More Patterns	Students pooled all Lesson 1 bottle-cap flip data (approx. 900 total trials) onto a shared class tally chart and constructed a live cumulative relative-frequency graph on the board, plotting how the fraction of 'cap-up' results changed as more trials were added. They also plotted coin-flip data on the same graph for comparison, producing two visible convergence lines: bottle-cap settling near 0.3 and coin settling near 0.5.	Students formally learned that experimental probability is defined as $P(E) = \text{number of favourable outcomes} \div \text{total number of trials}$. They learned that relative frequency converges toward a stable value as the number of trials increases (the Law of Large Numbers at an intuitive level), that equally likely outcomes (fair coin, $P = 0.5$) differ from unequally likely outcomes (bottle-cap, $P \approx 0.3$), and that larger samples produce more reliable probability estimates.	Students applied the $P(E)$ formula to Mama Njeri's three-month notebook, calculating $P(\text{good day}) = 24 \div 60 = 0.4$. They explained that this means approximately 40% of days are 'good days' over many days of trading — not that any specific tomorrow is guaranteed. They connected the convergence pattern in their bottle-cap graph to why Mama Njeri's fraction of good days stabilised over three months even though individual days remained unpredictable, advancing the class's answer to the driving question.

Lesson 3 Probability Space: Sample Spaces and Theoretical Probability	We observed that when we list all outcomes of an experiment systematically — using a list, a two-way table, or a grid — we always get the same total count of outcomes regardless of representation. We also observed that the theoretical probability of getting heads on a fair coin (0.5) matched the value our experimental relative frequency was converging toward in Lesson 2.	We learned that the sample space is the set of all possible outcomes of an experiment, that theoretical probability $P(E)$ equals the number of favourable outcomes divided by the total number of outcomes in the sample space, and that probability is always between 0 and 1 inclusive. We learned that combined events such as tossing two coins require organised tools like two-way tables to avoid missing or repeating outcomes.	The stabilisation of relative frequency in Lessons 1 and 2 is explained by theoretical probability: as the number of trials grows, the experimental proportion gets closer to the true fraction of favourable outcomes within the sample space. Mama Njeri experiences predictable long-run patterns precisely because there is an underlying probability space governing her sales outcomes, even though individual days remain uncertain.
Lesson 4 Mutually Exclusive Events: When Two Things Cannot Happen at Once	Learners sorted event-pair cards from Mama Njeri's market and Kenyan daily life into 'can happen at the same time' vs. 'cannot happen at the same time,' then physically placed outcome cards onto large Venn diagram templates to see which pairs produce an empty intersection.	Two events are mutually exclusive if they cannot both occur in the same trial — their intersection is empty ($A \cap B = \emptyset$, $P(A \cap B) = 0$). For mutually exclusive events, $P(A \cup B) = P(A) + P(B)$. Mama Njeri's 'sold out' and 'half unsold' outcomes are mutually exclusive, so their probabilities can be added directly.	Learners formalised the addition rule through a guided class discussion anchored in Mama Njeri's sales data, practised with tiered problems from Kenyan market, sports, and weather contexts, and updated their cumulative probability models with a specific quantitative recommendation for Mama Njeri's daily stocking decision.
Lesson 5 Independent Events: Does One Outcome Affect the Next?	Students tossed two coins simultaneously 40 times per group and recorded outcomes (HH, HT, TH, TT), then analysed printed Eldoret seed germination data cards showing whether paired seeds germinated. They tallied joint frequencies and computed observed relative frequencies for each outcome across both activities.	Students learned that two events are independent if the occurrence of one does not affect the probability of the other, formally defined as $P(A \cap B) = P(A) \times P(B)$. They verified this rule against their coin-toss and seed germination data. They also learned that independent events and mutually exclusive events are distinct concepts — mutually exclusive events cannot both occur, while independent events can both occur but do not influence each other.	Students explained why Mama Njeri's daily mandazi sales appear to behave as independent events — yesterday's outcome does not change the probability of today's outcome. Using the multiplication rule and relative frequency data from Lesson 4, they calculated the probability of two consecutive good sales days, and revised their cumulative probability models to reflect this understanding. They wrote a two-sentence evidence-based advice note to Mama Njeri applying the concept of independence to her daily production planning.
Lesson 6 The Addition Law of Probability — Overlapping Events in Mama Njeri's World	Students sorted Nakuru weather and sales data cards into four Venn diagram regions (HIGH SALES only, HEAVY RAIN only, BOTH, NEITHER) and counted $n(H \cup R) = 12$ out of 20 days, observing that simply adding $n(H) + n(R) = 15$ overcounts by exactly the 3 days in the intersection.	Students learned the Addition Law of Probability: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, recognised that mutually exclusive events are the special case where $P(A \cap B) = 0$ (reducing the law to the rule from Lesson 4), and applied the law to calculate that on approximately 60% of days, Mama Njeri can expect HIGH SALES or HEAVY RAIN (or both) — directly informing her daily mandazi production decision.	The double-counting problem explains why we must subtract $P(A \cap B)$: days that belong to BOTH events are included once in $P(A)$ and once in $P(B)$, so adding them directly inflates the probability. The Venn diagram makes this visible — the intersection region is counted in both circles, and subtracting it once gives the true count of days in the union. This is why Mama Njeri cannot simply add separate probabilities when events overlap in her real sales world.

Lesson 7 Multiplication Law of Probability: Independent and Dependent Events	Students observed (via tree diagram construction) that when maize seeds are drawn without replacement, the denominator of the second branch CHANGES to reflect the reduced sample space — the two events are LINKED. In contrast, when flipping a coin twice, the second branch denominators remain unchanged — the events are INDEPENDENT. Students observed that pathway probabilities on a tree diagram are calculated by multiplying the fractions along each branch, and that all pathway probabilities sum to 1.	Students learned that: (1) When two events are INDEPENDENT, $P(A \cap B) = P(A) \times P(B)$; (2) When two events are DEPENDENT (such as drawing without replacement), $P(A \cap B) = P(A) \times P(B A)$, where $P(B A)$ reads as 'the probability of B given that A has already occurred'; (3) The key question for determining which formula to use is: 'Does the outcome of the first event change the sample space for the second event?'; (4) For the maize seed scenario (3 germinating, 2 not, from 5), $P(\text{both germinate}) = 3/5 \times 2/4 = 6/20 = 3/10$.	Students explained (and can now use to advise Mama Njeri) that: mathematics can calculate the probability of TWO uncertain events BOTH occurring, by multiplying their individual probabilities — but only if you carefully check whether the events affect each other. This is more powerful than single-event probability because Mama Njeri's real business decisions often depend on MULTIPLE things going right at the same time (e.g., good customer turnout AND quality flour supply). The multiplication law gives her — and us — a precise mathematical tool to quantify those compound uncertainties, moving us closer to answering the driving question of how mathematics can support smarter decisions even when no outcome is certain.
Lesson 8 Probability Tree Diagrams: Mapping Sequential Outcomes	Students observed the teacher live-construct a two-stage probability tree diagram for Mama Njeri's consecutive good/bad day scenario, watching how probabilities are written on branches, how terminal path probabilities are found by multiplying along each branch-path, and how the probabilities of multi-path events are found by adding across relevant paths. Students also observed that all terminal path probabilities in the complete tree sum to 1.	Students learned that a probability tree diagram is a visual tool for systematically mapping all possible outcomes of a sequential (multi-stage) experiment; that the probability of a specific path through the tree equals the product of all branch probabilities along that path; that the probability of an event occurring via more than one path equals the sum of those path probabilities; and that this tool applies to both independent events (e.g., Mama Njeri's consecutive days, Kiambu seed germination) and dependent events with conditional branch probabilities (e.g., Kamau's matatu route and arrival time).	Students explained how Mama Njeri can use a two-day tree diagram to calculate the probability of two consecutive good days ($P = 0.16$), exactly one good day ($P = 0.48$), and two bad days ($P = 0.36$), and connected this to a concrete decision: because the most likely two-day outcome is exactly one good day (probability 0.48), Mama Njeri should plan her stock levels for a mix of good and bad days rather than assuming two good days in a row. Students also explained that tree diagrams make the multiplication rule for combined events visual and systematic, reducing errors in complex multi-stage calculations.
Lesson 9 Synthesis & Unit Assessment: Helping Mama Njeri	Students examined Mama Njeri's complete 60-day sales frequency table and noticed that roughly 40% of days resulted in sales of $\geq 80\%$ of stock, while the proportion of very poor days (less than 60% sold) was consistently small — a pattern that held even as individual daily outcomes varied widely.	Students applied the full probability toolkit: computing experimental probabilities from frequency data, constructing the sample space for a day's sales outcomes, classifying events on the 0-to-1 scale, using the Addition Law $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ for non-mutually-exclusive events, applying the Multiplication Law for independent consecutive days, and building a two-day tree diagram — all synthesised into a justified recommendation for Mama Njeri's optimal daily batch size.	The fraction of 'good days' stabilises over time because, although each individual day's sales outcome is uncertain (a random experiment), the relative frequency of favourable outcomes converges toward the true probability as the number of trials increases — this is the Law of Large Numbers. Probability does not eliminate uncertainty for any single day, but it gives Mama Njeri a mathematically grounded strategy: by trying a batch size calibrated to her experimental probability of high sales, she minimises expected losses across

			many days, even though she can never be certain about tomorrow.
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